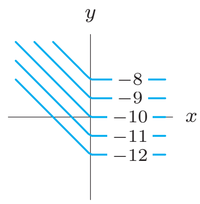
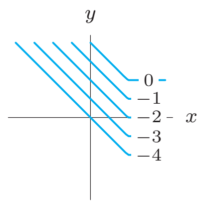
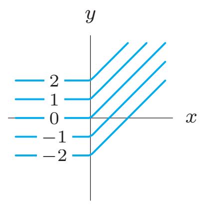
(b)



(c)



(d)



43 Contour diagram is curves in xy-plane

45 $f(x, y) = x^2$

47 Could not be true

49 Might be true

51 True

53 True

55 False

57 False

59 True

Section 12.4

1 -1.0

3 Not a linear function

5 Linear function

7 Linear

9 $z = \frac{4}{3}x - \frac{1}{2}y$

11 $z = -2y + 2$

13 $\Delta z = 0.4$; $z = 2.4$

15 CDs cost \$8
DVDs cost \$12

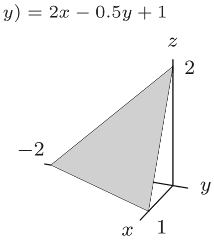
17 (a) Linear
(b) Linear
(c) Not linear

19 180 lb person at 8 mph
120 lb person at 10 mph

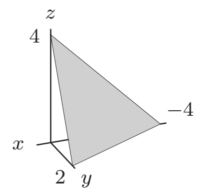
21 $g(x, y) = 3x + y$

23 $f(x, y) = 2x - 0.5y + 1$

25



27



31 (a) $7/\sqrt{29}$
(b) $-5/\sqrt{104}$

33 Contours of $f(x, y) = e^{x+y}$ parallel lines

35

$x\backslash y$	1	2	3
1	1	2	3
2	2	4	6
3	3	6	9

37 False

39 False

41 True

43 True

45 True

47 False

Section 12.5

1 (a) I
(b) II

3 $f(x, y) = \frac{1}{3}(5 - x - 2y)$

5 $f(x, y) = (1 - x^2 - y)^2$

7 Elliptical and hyperbolic paraboloid, plane

9 Hyperboloid of two sheets

11 Ellipsoid

13 Yes, $f(x, y) = (2x + 3y - 10)/5$

15 No

17 $f(x, y) = \sqrt{10 - x^2 - y^2}$
$g(x, y, z) = x^2 + y^2 + z^2 = 10$

19 (a) Parallel planes: $2x - 3y + z = c + 20$
(b) $f_z(0, 0, 0) = 1$
(c) $\vec{n} = 2\vec{i} - 3\vec{j} + \vec{k}$
(d) Yes; $-3°$F

21 Incr of A and r
Decr of t

23 $f(x, y) = 3\sqrt{1 - x^2 - y^2/4}$;
$g(x, y) = -3\sqrt{1 - x^2 - y^2/4}$

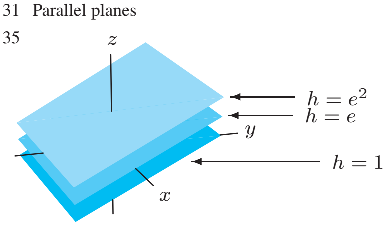
25 (a) Graph of f is graph of
$x^2 + y^2 + z^2 = 1, z \geq 0$
(b) $\sqrt{1 - x^2 - y^2} - z = 0$

27 $g(x, y, z) = 1 - \sqrt{x^2 + z^2} - y = 0$

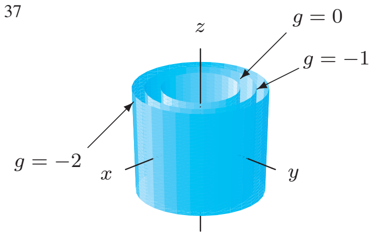
29 Ellipsoid

31 Parallel planes

35



37



39 Level surfaces hyperbolic cylinders

41 $f(x, y, z) = y + z$

43 $f(x, y, z) = (x + y + z)^2$

45 True

47 True

49 True

51 False

53 False

Section 12.6

1 Not continuous

3 Continuous

5 Not continuous

7 1

9 0

11 1

19 No

21 $c = 1$

23 (c) No

25 For quotient, need $g(a, b) \neq 0$

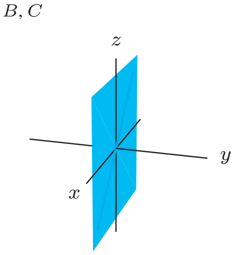
27 $f(x, y) = (x^2 + 2y^2)/(x^2 + y^2)$

29 $f(x, y) = 1/((x - 2)^2 + y^2)$

Chapter 12 Review

1 A, B, C

3



5 Not a function

7 IV

9 (a) is (I)
(b) is (IV)
(c) is (II)
(d) is (III)

11 $y = k$